Gozde KARATAS BAYDOGMUS

Assistant Professor | AI and Machine Learning Research Fellow Loyola University Chicago

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Professional Summary

Ph.D. in Computer Engineering specializing in federated learning, privacy-preserving AI, trustworthy machine learning, and cybersecurity applications, with emerging research in healthcare AI. Academic leader and researcher with experience supervising doctoral and master's students, mentoring interdisciplinary research teams, securing competitive research funding, and teaching AI, security, programming, and data science at undergraduate and graduate levels. Former Vice Head of Department at two universities with a strong record of research, student mentorship, and curriculum development.

Education

Marmara University — Istanbul, Turkey
Ph.D., Computer Engineering — GPA: 3.84/4.00 — *Sept. 2015 – May 2020*
- Thesis: *Intrusion Detection System Using Deep Learning*

Istanbul Kultur University — Istanbul, Turkey
M.S., Computer Engineering — GPA: 3.86/4.00 — *Sept. 2013 – May 2015*
- Thesis: *Testing Utilities For Distributed Database's*

Istanbul Kultur University — Istanbul, Turkey
B.S., Mathematics and Computer Sciences — GPA: 3.83/4.00 — *Sept. 2009 – May 2013*

Academic Positions

Loyola University Chicago — Chicago, USA
Research Fellow - Visiting Scholar, Dept. of Computer Science — *2025 – Present*
- Conducting research in machine learning and AI systems
- Mentoring doctoral and undergraduate students; collaborating on grant-funded research and manuscripts

Marmara University — Istanbul, Turkey
Assistant Professor & Vice Head of Department, Computer Engineering — *2022 – 2025*
- Designed and validated AI/ML models through systematic benchmarking and reproducible experimentation
- Supervised doctoral and master's theses in federated learning, intrusion detection, malware analysis, and computer vision

Biruni University — Istanbul, Turkey
Assistant Professor & Vice Head of Department, Computer Engineering — *2021 – 2022*
- Published scientific articles and presented research at international conferences
- Provided academic and industry consultancy on AI and cybersecurity projects

Istanbul Kultur University — Istanbul, Turkey
Research Assistant & SAP Faculty Coordinator, Dept. of Mathematics and Computer Science — *2014 – 2021*
- Conducted research in machine learning, intrusion detection, and distributed database systems
- Coordinated SAP faculty activities; progressed from teaching assistant to course instructor

Professional Activities

AI/ML/Cybersecurity Freelance Consultant — Remote
Independent — *2021 – Present*
- Provided technical consultancy to industry clients on AI, cybersecurity, and ML system design

University of Oviedo — Oviedo, Spain
Visiting Academic (Erasmus Staff Mobility) — *2024*
- Activities included academic collaboration, departmental meetings, and exchange of teaching/research practices.

Grants & Funding

Funded / Completed

Forensic Detection and Applicability Assessment of Digital Image Forgery | 2026 – Cont.

- *Marmara University Scientific Research Project (BAPKO)*
- Forensic detection of digital image forgery using transfer-learning-based deep learning models
- Requested Amount: 150,000 TL

Emergent Locomotor Agility – Multi-Agent Adversarial Games | 2025 – 2026

- *Loyola University Chicago ORS Grant*
- Adversarial self-play and goal-conditioned RL to induce emergent agility across randomized terrains and obstacles
- Budget: $5,000

Travel Path Planning Using Ant Colony Optimization | 2022 – 2024

- *The Scientific and Technological Research Council of Turkey (TÜBİTAK)*
- Path optimization algorithm for city Amasya; improved parallel clustering and elitist ACO for TSP/mTSP
- Budget: 79,000 TL

Data Privacy and Security Methods in Federated Learning Systems | 2025

- *The Scientific and Technological Research Council of Turkey (TÜBİTAK), Program 2209-A – Undergraduate Research Projects Support Program*
- Advisor for undergraduate research project (Student: Özer Öztürk)
- Budget: 12,000 TL

Model Training and Real-World Analysis Using Health Data with Federated Learning | 2025

- *The Scientific and Technological Research Council of Turkey (TÜBİTAK), Program 2209-A – Undergraduate Research Projects Support Program*
- Advisor for undergraduate research project (Students: Abdülbaki Demir, Ahmet Yasir Kulaksız)
- Budget: 12,000 TL

Developing an Effective FL-Based Defense System Against Phishing Attacks on Mobile Platforms | 2025

- *The Scientific and Technological Research Council of Turkey (TÜBİTAK), Program 2209-A – Undergraduate Research Projects Support Program*
- Advisor for undergraduate research project (Student: Eren Doğan)
- Budget: 12,000 TL

Network IDS Using Voting-Based Ensemble Learning | 2020 – 2022

- *Marmara University Scientific Research Project (BAPKO)*
- Voting-based ensemble classifier for network intrusion detection; 95%+ accuracy on benchmark datasets
- Budget: 90,000 TL

Deep Learning Based Intrusion Detection System | 2018 – 2020

- *Marmara University Scientific Research Project (BAPKO)*
- RNN/LSTM classifiers outperforming conventional IDS methods; SHAP-based interpretability analysis
- Budget: 150,000 TL

Research Projects (2025-2026)

AI Agent-Based Financial Fraud Detection System | 2025

- *Biruni University, Undergraduate Project*
- 135-feature behavioral pipeline (velocity, fraud history, amount z-scores) on the IEEE-CIS Fraud Detection dataset; LR → XGBoost/LightGBM (Optuna TPE) → ensemble comparison
- ReAct-style autonomous agent combining ML score, SHAP explanations, and rule-based velocity checks to issue APPROVE/REVIEW/FLAG decisions with human-readable justifications

FastStroke: Hybrid Clinical Decision Support System for Facial Palsy | 2025

- *Biruni University, Undergraduate Project*

- Two-layer hybrid architecture: YOLO26n-cls transfer learning (LOPO protocol) combined with MediaPipe Face Mesh landmarks and an RBF-SVM on MAR/EAR features
- Addressed temporal leakage and domain shift on the YFP dataset; age-normalized 0–100 risk score; deployed as a Flask/SQLite clinical web application

Learning to Rank for E-Commerce Product Search | 2025
- *Biruni University, Undergraduate Project*
- Fully automated, zero-annotation LTR pipeline on Amazon Reviews 2023; rule-based, LLM-based (GPT-4o), and hybrid synthetic query generation
- Snorkel-inspired weak supervision and a two-stage dense retrieval (MiniLM) + LightGBM LambdaRank reranking architecture

AI Agent-Assisted Liver Cirrhosis Histological Stage Classification | 2025
- *Biruni University, Undergraduate Project*
- Multi-class staging on the Mayo Clinic PBC dataset (9,639 records after deduplication); 5-way ensemble comparison (AdaBoost, Bagging, Voting, Stacking)
- Best stacking model (XGBoost + RF + LightGBM, LR meta-learner) deployed via an interactive Streamlit scenario engine for clinical what-if analysis

Invited Talks

"Federated Unlearning for Biometric Recognition" September 2026
- *WiCyS Tech Research Mixer, Loyola University Chicago*

"From Privacy to Forgetting" November 2025
- *CS Department Fall 2025 Seminar Series, Loyola University Chicago*

Publications

Journal Articles

Baydogmus Karatas, G., & Demir, O. (2026). Balancing the Truth: Social Media Misinformation Detection via GraphSAGE and Feature Diversity. *PeerJ Computer Science*, 12. doi:10.7717/peerj-cs.3975 – *Q2, JIF 2.5*

Buyuktanir, B., **Baydogmus Karatas, G.**, Ulku, E. E., & Yildiz, K. (2026). Enhancing employee retention prediction through federated learning and associative classification using FedCBA: A case study on IBM HR analytics. *Arabian Journal for Science and Engineering*. doi:10.1007/s13369-026-11252-8 – *Springer*

Buyuktanir, B., Kucur, E., Kuyumcu, Z., Ozdemir, N., Altinkaya, S., **Baydogmus Karatas, G.** (2026). Machine unlearning through model forgetting techniques: application area and emerging methods. *Cluster Computing*, 29(9), 579.

Sahingoz, O. K., **Baydogmus Karatas, G.**, & Kugu, E. (2026). A systematic literature review of You Only Look Once architectures (v1–v12) in healthcare systems. *Diagnostics*, 16(6), 935. doi:10.3390/diagnostics16060935

Kurt, A., Cakir, A., Polat, C.C., Busra, B., **Baydogmus Karatas, G.**, & Yildiz, K. (2026). Evaluating SISA-Based Machine Unlearning Across Diverse Modalities: Tabular, Visual and Auditory Data. *Journal of Naval Sciences and Engineering*, 151–187. doi:10.56850/jnse.1829992

Demir, A., Kulaksiz, A. Y., Buyuktanir, B., **Baydogmus Karatas, G.**, & Yildiz, K. (2026). Decentralized disease prediction: A federated learning perspective. *Afyon Kocatepe University Journal of Science and Engineering*, 26(2), 337–353. doi:10.35414/akufemubid.1679417

Buyuktanir, B., Altinkaya, S., **Baydogmus Karatas, G.**, & Yildiz, K. (2025). Federated learning in intrusion detection: advancements, applications, and future directions. *Cluster Computing*, 28(473). doi:10.1007/s10586-025-05325-w – *Springer*

Baydogmus, G. K., & Demir, O. (2025). Awareness with Machine: Hybrid Approach to Detecting ASD with a Clustering. *Computers, Materials & Continua*, 84(2). doi:10.32604/cmc.2025.062643

Altinel, A. B., Sahin, S., Gurbuz, M. Z., & **Baydogmus, G. K.** (2024). So-haTRed: A novel hybrid system for Turkish Hate Speech Detection in Social Media with Ensemble Deep Learning improved by BERT and clustered-Graph Networks. *IEEE Access*. doi:10.1109/ACCESS.2024.3415350 – *Q2, JIF 3.6*

Emanet, S., **Baydogmus Karatas, G.**, & Demir, O. (2023). An ensemble learning based IDS using Voting rule: VEL-IDS. *PeerJ Computer Science*, 9, e1553. doi:10.7717/peerj-cs.1553 – *Q2, JIF 2.5*

Baydogmus Karatas, G. (2023). Solution for TSP/mTSP with an improved parallel clustering and elitist ACO. *Computer Science and Information Systems*, 20(1), 195–214.

Emanet, S., **Karatas Baydogmus, G.**, & Demir, O. (2021). Effects of Feature Selection Methods on Machine Learning Based Intrusion Detection System Performance. *Dicle University Journal of Engineering*, 12(5).

Karatas Baydogmus, G. (2021). Detecting Internet of Things Attacks by Using Hybrid Learning and Feature Selection Method. *European Journal of Science and Technology*, 29, 19–25.

Karatas Baydogmus, G. (2021). The effects of normalization and standardization on Internet of Things attack detection. *European Journal of Science and Technology*, 29, 187–192.

Karatas, G., Demir, O., & Sahingoz, O. K. (2020). Increasing the Performance of Machine Learning-Based IDSs on an Imbalanced and Up-to-Date Dataset. *IEEE Access*, 8, 32150–32162. doi:10.1109/ACCESS.2020.2973219 – *Q2, JIF 3.6*

Sahingoz, O. K., Cebi, C. B., Bulut, F. S., Firat, H., & **Karatas, G.** (2020). Deep learning based security management of information systems: A comparative study. *Journal of Advances in Information Technology*, 11(3), 135–142. doi:10.12720/jait.11.3.135-142

Sahingoz, O. K., Cebi, C. B., Bulut, F. S., Firat, H., & **Karatas, G.** (2019). Comparison of Machine Learning Based Models in Intrusion Detection Systems. *Erzincan University Journal of Science and Technology*, 12(3), 1513–1525.

Karatas, G., & Akbulut, A. (2018). Survey on access control mechanisms in cloud computing. *Journal of Cyber Security and Mobility*, 7(3), 1–36. – *River Publishers*

Akbulut, A., & **Karatas, G.** (2018). Automated testing for distributed databases with fuzzy fragment reallocation. *Turkish Journal of Electrical Engineering & Computer Sciences*, 26(2), 677–692. doi:10.3906/elk-1701-1350

Karatas, G., Akbulut, A., & Zaim, A. H. (2016). Security In Mobile Devices - Threats And Basic Strategies. *Istanbul Commerce University Journal of Science*, 15(30), 55–75.

Conference Papers

Onlu, A. O., Akca, B., Buyuktanir, B., Yildiz, K., & **Baydogmus Karatas, G.** (2025). An investigation and performance evaluation of aggregation algorithms in federated learning architecture. *2025 Innovations in Intelligent Systems and Applications Conference (ASYU)*, pp. 1–6. IEEE.

Tosun, B. E., Yorulmaz, E., Ergul, F. E., Buyuktanir, B., **Baydogmus Karatas, G.**, & Yildiz, K. (2025). Comparative analysis of relational classification algorithms on benchmark datasets. *2025 10th International Conference on Computer Science and Engineering (UBMK)*, pp. 790–795. IEEE.

Buyuktanir, B., Yildiz, K., & **Baydogmus Karatas, G.** (2025). A Systematic Mapping Study on Machine Unlearning in Federated Learning. *2025 7th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA)*, pp. 1–10. IEEE.

Ozdemir, S. N., **Baydogmus Karatas, G.**, & Demir, O. (2025). CNN and LSTM-Based In-Memory Malware Detection: Performance Improvement with Transfer Learning. *2025 7th ICHORA*, pp. 1–6. IEEE.

Demir, A., Kulaksiz, A. Y., Buyuktanir, B., **Baydogmus Karatas, G.**, & Yildiz, K. (2025). Model Training and Real World Analysis Using Health Data with Federated Learning. *International Open Source Conference (UAKK) 2024*.

Karci, S., & **Karatas Baydogmus, G.** (2024). Obtaining the Dominating Set in Graphs with the Invasive Weed Optimization Algorithm. *2024 8th International Artificial Intelligence and Data Processing Symposium (IDAP)*, pp. 1–5. IEEE.

Cicekli, N. Z., **Karatas Baydogmus, G.**, & Altinkaya, S. (2024). Exploring Machine Learning Techniques For Gender Voice Recognition Using Limited Speech Data. *2024 8th International Conference on Computational Mathematics and Engineering Sciences*.

Karatas Baydogmus, G., & Altinkaya, S. (2024). A Survey On Different Statistical Distributions Using Python Programming. *2024 8th International Conference on Computational Mathematics and Engineering Sciences*.

Baydogmus Karatas, G., & Cicekli, N. Z. (2023). Disease Detection using Deep Learning Algorithms on the Hardware Platforms. *2023 Innovations in Intelligent Systems and Applications Conference (ASYU)*, pp. 1–6. IEEE.

Cicekli, N. Z., & **Baydogmus Karatas, G.** (2022). Covid-19 disease detection with improved deep learning algorithms on x-ray data. *2022 HORA*, pp. 1–4. IEEE.

Kirci, B. K., & **Baydogmus Karatas, G.** (2022). The Effect of Loss and Optimization Functions on Bitcoin Rate Prediction in LSTM. *2022 HORA*, pp. 1–4. IEEE.

Baydogmus Karatas, G. (2022). Probability Selection for Solving Sudoku with Ant Colony Optimization Algorithm. *2022 1st International Conference on Information System & Information Technology (ICISIT)*, pp. 161–165. IEEE.

Baydogmus Karatas, G. (2022). A Parallelization Based Ant Colony Optimization for Travelling Salesman Problem. *2022 1st ICISIT*, pp. 166–169. IEEE.

Karatas, G., Demir, O., & Sahingoz, O. K. (2019). A Deep Learning Based Intrusion Detection System on GPUs. *2019 11th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*, pp. 1–6. IEEE.

Osken, S., Yildirim, E. N., **Karatas, G.**, & Cuhaci, L. (2019). Intrusion detection systems with deep learning: A systematic mapping study. *2019 Scientific Meeting on Electrical-Electronics & Biomedical Engineering and Computer Science (EBBT)*, pp. 1–4. IEEE.

Karatas, G., Demir, O., & Sahingoz, O. K. (2018). Deep learning in intrusion detection systems. *2018 International Congress on Big Data, Deep Learning and Fighting Cyber Terrorism (IBIGDELFT)*, pp. 113–116. IEEE.

Reis, B., Kaya, S. B., **Karatas, G.**, & Sahingoz, O. K. (2018). Intrusion Detection Systems with GPU-Accelerated Deep Neural Networks and Effect of the Depth. *2018 6th International Conference on Control Engineering & Information Technology (CEIT)*, pp. 1–8. IEEE.

Karatas, G., & Sahingoz, O. K. (2018). Neural network based intrusion detection systems with different training functions. *2018 6th International Symposium on Digital Forensic and Security (ISDFS)*, pp. 1–6. IEEE.

Karatas, G., Can, F., Dogan, G., Konca, C., & Akbulut, A. (2017). Multi-tenant architectures in the cloud: A systematic mapping study. *2017 International Artificial Intelligence and Data Processing Symposium (IDAP)*, pp. 1–4. IEEE.

Catal, C., **Karatas, G.**, et al. (2016). A Systematic Mapping Study on Software Architecture Recovery. *International Conference on Computer Science and Engineering (UBMK 2016).*

Karatas, G. (2016). Genetic algorithm for intrusion detection system. *2016 24th Signal Processing and Communication Application Conference (SIU)*, pp. 1341–1344. IEEE.

Karatas, G., & Catal, C., et al. (2014). Yalin Yazilim Gelistirme Sistematik Esleme Calismasi. *Ulusal Yazilim Muhendisligi Sempozyumu (UYMS 14).*

Preprints / Other Publications

Karatas, C., Karadogan, H., Ertug, A. Y., Buyuktanir, B., Yildiz, K., & **Baydogmus, G. K.** (2026). Federated Learning Architecture: Data Privacy and System Security Approaches. *arXiv preprint arXiv:2607.09391.*

Buyuk, F. B., **Karatas Baydogmus, G.**, Buldu, A., Tulendiyeva, A., & Baizhumanova, Z. (2026). Digital Image Forgery Detection Using Transfer Learning. *arXiv preprint arXiv:2605.08167.*

Ozturk, O., Buyuktanir, B., **Baydogmus, G. K.**, & Yildiz, K. (2025). Differential Privacy in Federated Learning: Mitigating Inference Attacks with Randomized Response. *arXiv:2509.13987.*

Yenigun, O., **Baydogmus, G. K.**, & Yildiz, K. (2025). Smart Routing for EV Charge Point Operators in Mega Cities: Case Study of Istanbul. *arXiv:2509.21369.*

Teaching Experience

Marmara University, Dept. of Computer Engineering – *Instructor* Istanbul, Turkey, 2022–2024

Automata Theory (BLM3059) – undergraduate
Computer Security (BLM3060) – undergraduate
Cryptography (BLM4062) – undergraduate
Computer Programming with Python (BLM2001) – undergraduate
Data Structures (BLM2002) – undergraduate
Computer Security and Cryptography (BLM7003) – graduate
Artificial Intelligence (BLM7035) – graduate

Biruni University, Dept. of Computer Engineering – *Instructor* Istanbul, Turkey, 2021–2022

Object Oriented Programming (BLG201) – undergraduate

Istanbul Kultur University, Dept. of Mathematics and Computer Sciences Istanbul, Turkey, 2014–2021

Computer Networks (MB0053) – Teaching Assistant (2018–2019), Instructor (2020–2021)
Data Mining (MB00538) – Teaching Assistant (2018–2019), Instructor (2020)
Algorithm Analysis and Design (MB6002) – Teaching Assistant (2014–2020)
Applied Statistics (MB0036) – Teaching Assistant (2014–2020)
Data Structures – Teaching Assistant (2014–2021)
Computer Programming with Python – Teaching Assistant (2014–2021)
Object Oriented Programming – Teaching Assistant (2014–2020)

Student Advising

Doctoral Students

Busra Buyuktanir – Marmara University (co-advised with K. Yildiz); *FedFORGET: Machine Unlearning Approach for Federated Learning Architecture*

Master's Students

Onur Yenigun – Marmara University (co-advised with K. Yildiz), defended 2025; *Route optimization for EV charge point operators using K-Means and Genetic Algorithm*

Seymanur Ozdemir – Marmara University (co-advised with O. Demir), defended 2025; *Malware detection with deep neural networks and transfer learning*

Tuba Koylu – Marmara University (co-advised with K. Yildiz); *Computer Vision-Based Damage Assessment of the 2023 Kahramanmaras Earthquakes: The Case of Antakya*

Sura Emanet – Marmara University (co-advised with O. Demir), defended 2022; *Collective machine learning based network intrusion detection*

Undergraduate Students (Conference Publications)

Batuhan Emre Tosun, Erdem Yorulmaz, Furkan Enes Ergul – UBMK 2025

Ahmet Onder Onlu, Burak Akca – ASYU 2025

Abdulbaki Demir, Ahmet Yasir Kulaksiz – UAKK 2024

Berke Kaan Kirci – HORA 2022

Nahide Zeynep Cicekli – HORA 2022

Academic Leadership

Departmental & University Service

Vice Head of Department, Computer Engineering, Marmara University (2022 – 2025) – Oversaw departmental operations, curriculum planning, and strategic decision-making

Vice Head of Department, Computer Engineering, Biruni University (2021 – 2022) – Supported departmental administration and academic program coordination

Reviewer Activities

Journals:
IEEE Access (Q2, JIF 3.6);
IEEE Transactions on Emerging Topics in Computational Intelligence (Q1, JIF 6.5);
IEEE Transactions on Dependable and Secure Computing (Q1, JIF 7.5);
IEEE Transactions on Network and Service Management (Q1, JIF 5.4);
IEEE Sensors Letters (Q3, JIF 2.2);
PeerJ Computer Science (Q2, JIF 2.5);
Digital Threats: Research and Practice (Q3, JIF 2.3);
Computational Intelligence (Q3, JIF 1.7);
IET Communications (Q3, JIF 1.6);
IET Information Security (Q2, JIF 2.6);
International Journal of Numerical Modelling (Q2, JIF 1.7)

Professional Memberships

ACM EIGTRUST – Emerging Interest Group on Trustworthy and Responsible Systems, Member, 2026–Present
PeerJ Computer Science – Lifetime Member

Certifications

Associate Data Scientist in Python
TensorFlow for Deep Learning Bootcamp
PyTorch for Deep Learning Bootcamp
Machine Learning: ML, DL, AI with AWS, Python
AWS Certified Cloud Practitioner

Continuing Education / Workshops

Anthropic Academy: Building with Claude, Prompt Engineering, AI Agents and Tool Use
DeepLearning.AI: Attention in Transformers: Concepts and Code in PyTorch (StatQuest); How Transformer LLMs Work (Jay Alammar, Maarten Grootendorst); Intro to Federated Learning (Flower Labs); How Diffusion Models Work; Agentic AI; Generative AI for Everyone; Federated Fine-tuning of LLMs with Private Data (Flower Labs); ChatGPT Prompt Engineering for Developers (OpenAI); LangChain for LLM Application Development
ACM Learning: "Inside an LLM Agent: A From-Scratch Walkthrough" (V. A. Fajardo, June 2026)

Technical Skills

Languages & Frameworks: Python, PyTorch, NVFlare, TensorFlow, scikit-learn, PySpark, C/C++, Java, JavaScript, MATLAB, SQL (MySQL, PostgreSQL), MongoDB
AI/ML: Deep Learning, Federated Learning, Machine Unlearning, Reinforcement Learning, MLOps, SHAP Analysis, Trustworthy AI, Statistical Modeling, Data Analysis, Reproducible AI, SQL, Docker, Git, Medical Imaging
Cybersecurity: Intrusion Detection Systems, Network Security, Malware Analysis, Cryptography, Privacy-Preserving ML
Tools: Docker, HuggingFace, TFDS, HDFS, Git